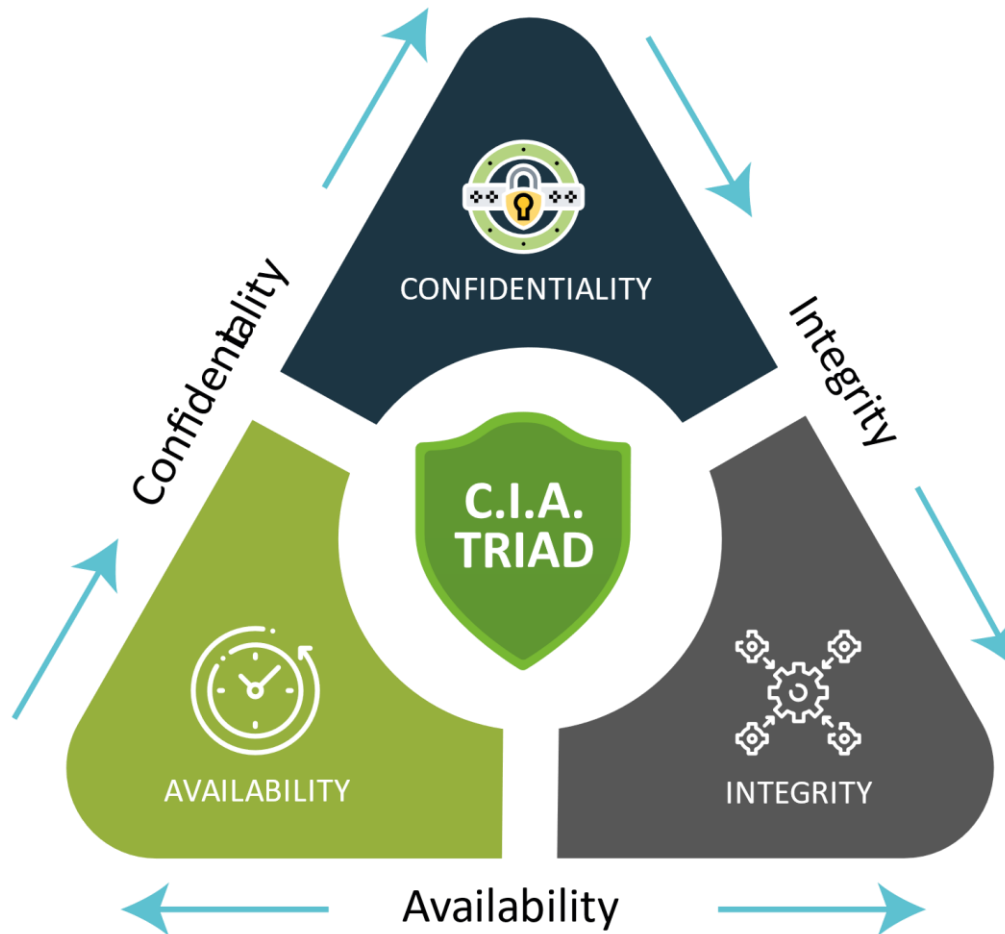


Computer Network Security Overview

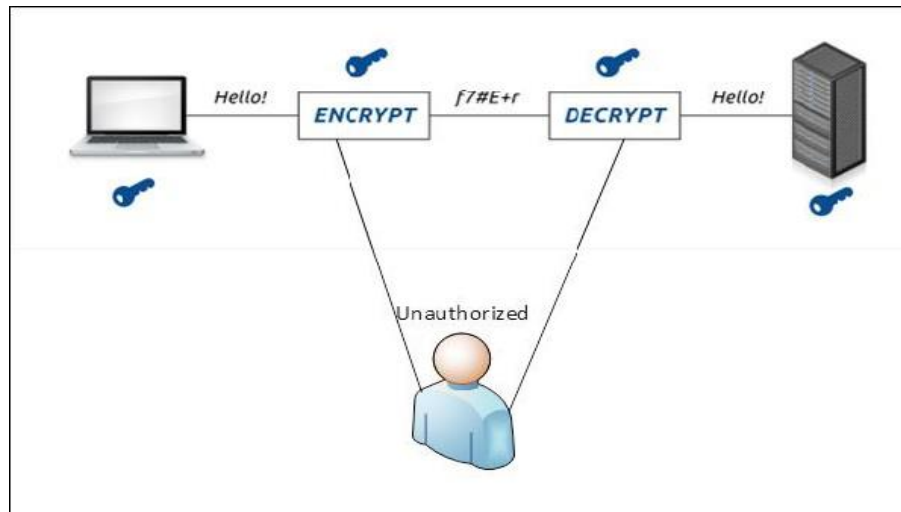


Different Elements in Computer Security



Confidentiality

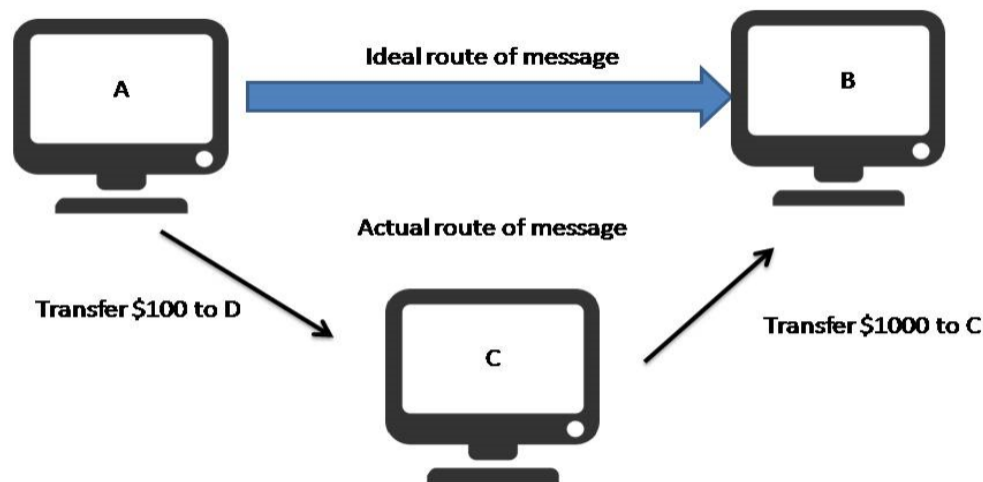
Confidentiality is the concealment (আড়াল) of information or resources. Also, there is a need to keep information secret from other third parties that want to have access to it, so **just the right people can access it.**



Example in real life – Let's say there are two people communicating via an encrypted email they know the decryption keys of each other and they read the email by entering these keys into the email program. If someone else can read these decryption keys when they are entered into the program, then the confidentiality of that email is compromised.

Integrity

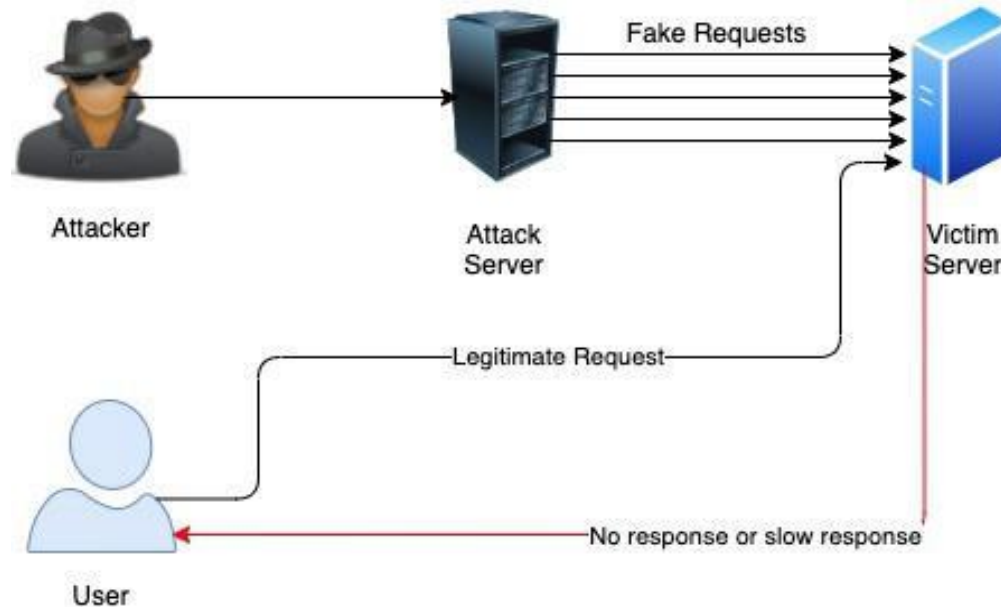
Integrity refers to the accuracy and completeness of data. Integrity involves maintaining the consistency and trustworthiness of data over its entire life cycle. **Data must not be changed in transit, and precautionary steps must be taken to ensure that data cannot be altered by unauthorized people.**



Example in real life – Let's say you are doing an online payment of 100 USD, but your information is tampered without your knowledge in a way by sending to the seller 1000 USD, this would cost you too much.

Availability

Availability means that information is accessible to authorized users. It provides an assurance that your system and data can be accessed by authenticated users whenever they're needed. Similar to confidentiality and integrity, availability also holds great value.



Example in real life – Let's say a hacker has compromised a webserver of a bank and put it down. You as an authenticated user want to do an e-banking transfer but it is impossible to access it, the undone transfer is a money lost for the bank.

Should an expert Use the CIA Triad?

The CIA triad provides a simple yet comprehensive high-level checklist for the evaluation of your security procedures and tools.

An effective system satisfies all three components: confidentiality, integrity, and availability. An information security system that is lacking in one of the three aspects of the CIA triad is insufficient.

The CIA security triad is also valuable in assessing what went wrong—and what worked—after a negative incident. For example, perhaps availability was compromised after a malware attack such as ransomware, but the systems in place were still able to maintain the confidentiality of important information. This data can be used to address weak points and replicate successful policies and implementations.

Confidentiality Attacks:

- Data breaches:** Unauthorized access to sensitive data due to vulnerabilities (e.g., poor encryption, phishing).
- Insider threats:** Employees or insiders with access abuse their privileges, often due to lack of strict access controls.

Integrity Attacks:

- Data manipulation:** Altering data without authorization, such as in cases of SQL injection or malware infections.
- Man-in-the-middle attacks:** An attacker intercepts and possibly alters communications between two parties.
- Ransomware:** Certain ransomware not only encrypts files but also alters them, affecting their original state.

Availability Attacks:

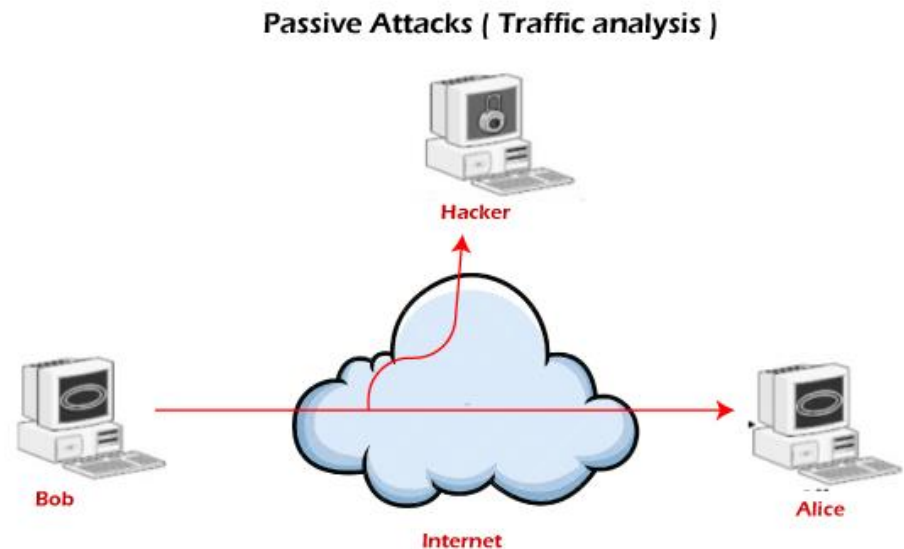
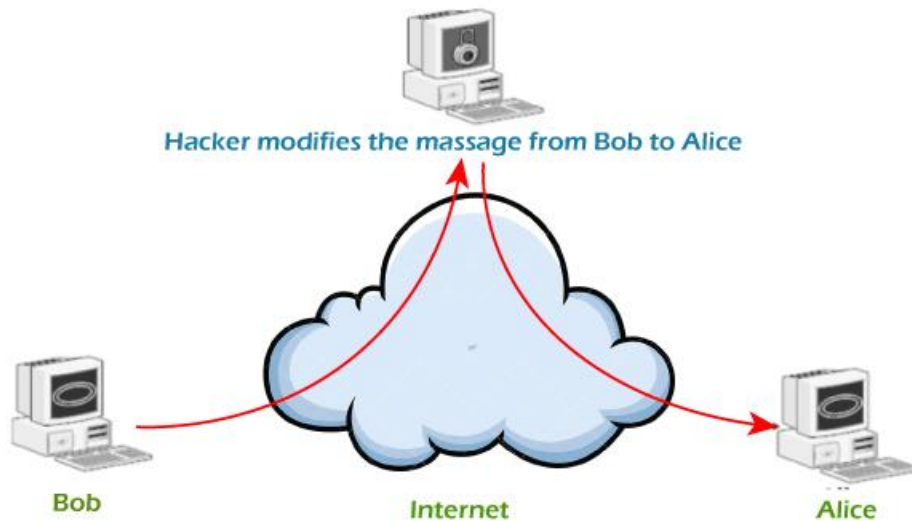
- Denial-of-Service (DoS) attacks:** Overwhelming a server or network to make resources unavailable.
- Hardware failures:** Lack of redundancy or backup can lead to loss of availability if critical hardware fails.
- Natural disasters:** Events like floods, earthquakes, or fires can disrupt services, which is why data centers often have disaster recovery plans.

Attacks

- Attacks on computer systems
 - break-in to destroy information
 - break-in to steal information
 - blocking to operate properly
 - malicious software
 - wide spectrum of problems

Attacks

- Network Security
 - Active attacks
 - Passive attacks

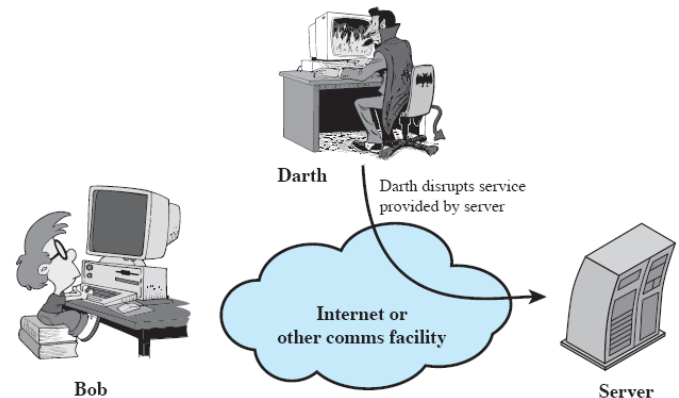
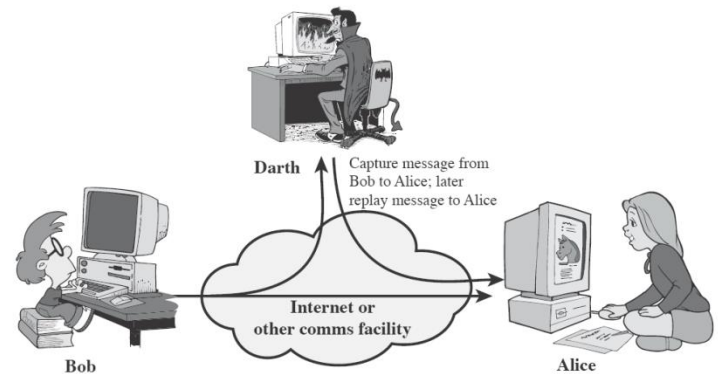
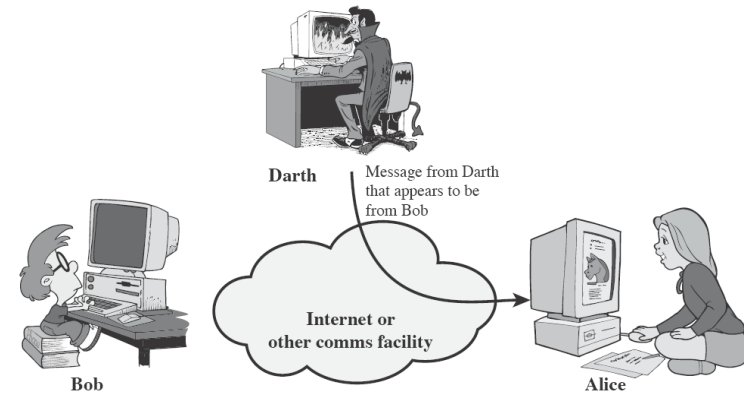


Attacks

- Passive attacks
 - An attacker observes the messages and copies them
 - victims do not get informed about the attack
 - It is difficult to detect as there is no alteration in the message.
 - Passive attacks can be prevented by using some **encryption techniques.**

Attacks

- Active attacks
 - Attacker actively manipulates the communication
 - Masquerade
 - pretend as someone else
 - possibly to get more privileges
 - Replay
 - passively capture data and send later
 - Denial-of-service
 - attempt to overwhelm a web property with traffic in order to disrupt its normal operations.



Security Mechanisms

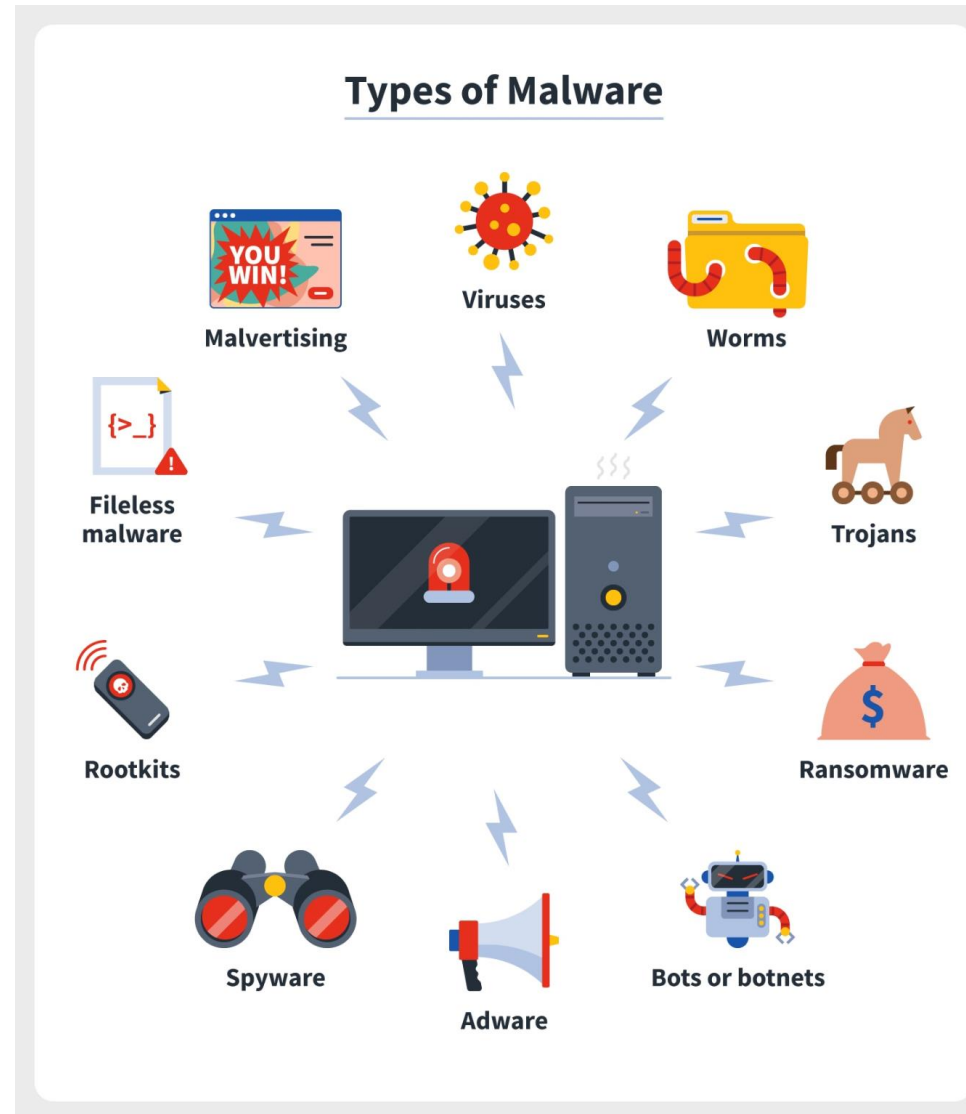
- Cryptographic Techniques
 - will see next
- Software and hardware for access limitations
 - Firewalls
- Traffic Padding
 - the process of creating fake data units, communication instances, or data within data units to hide the quantity of real data units being sent..
- Hardware for authentication
 - Smartcards, security tokens.
- Physical security
 - Keep it in a safe place with limited and authorized physical access

Common Types of Networking Attacks

- Malware
- Computer Virus
- Computer Worm
- Phishing
- DoS (Denial of Service) and DDoS
- Man-in-the-middle
- Ransomware
- SQL Injection Attacks

Malware

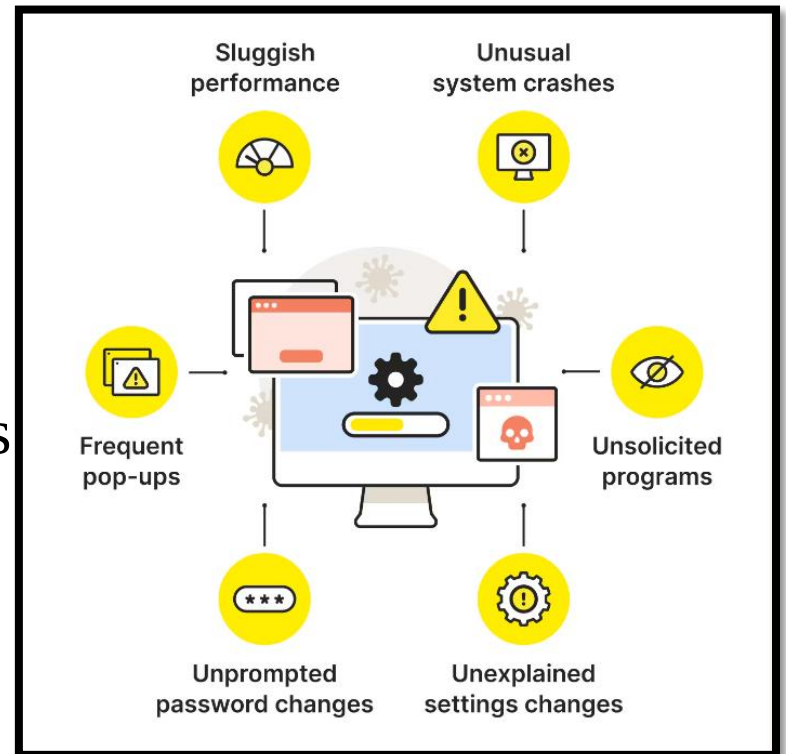
- Malware, short for malicious software, refers to any intrusive software developed by cybercriminals (often called hackers) to steal data and damage or destroy computers and computer systems. Examples of common malware include viruses, worms, Trojan viruses, spyware, adware, and ransomware.
- Hackers attempt to gain unauthorized access into the target system and disrupt or corrupt the files and data through malicious codes called malware.



Computer Virus

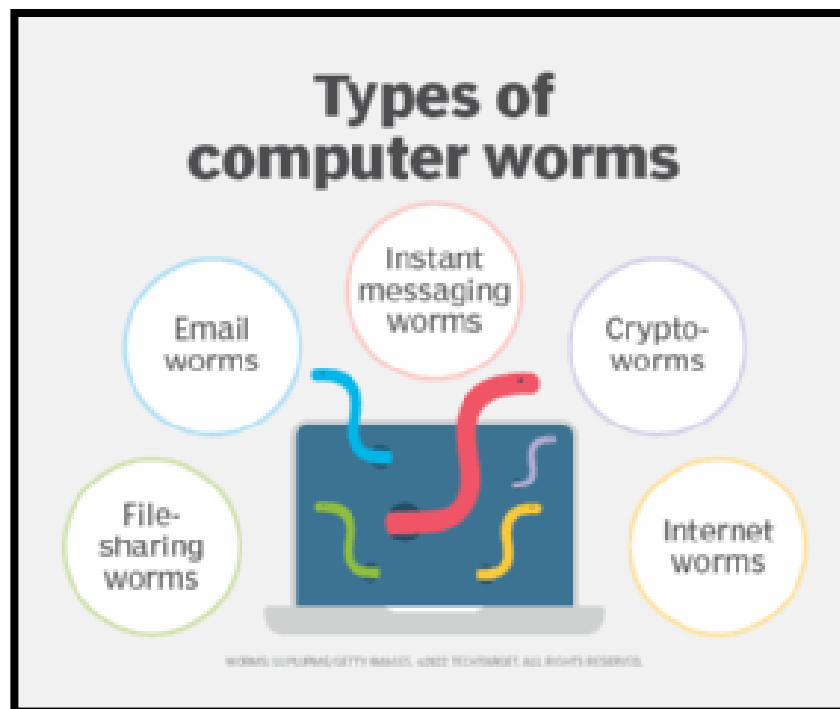
A type of malware, they are unique pieces of code that can wreak havoc(ধ্বংসযজ্ঞ) and spread from computer to computer.

If you click on an email with a malicious link or download links from infected websites, these viruses can corrupt your files, infect other computers from your list and steal your personal information.



Computer Worm

A computer worm is a type of malware whose primary function is to self-replicate and infect other computers while remaining active on infected systems. A computer worm duplicates itself to spread to uninfected computers.



Phishing

Phishing is a type of social engineering where an attacker sends a fraudulent(প্রতারণামূলক) (e.g., spoofed, fake, or otherwise deceptive) message designed to trick a person into revealing sensitive information to the attacker



From: [redacted] <[redacted]@MSVU.CA>

Sent: Friday, September 16, 2022 5:22 PM

Subject: We received a request from you

Our record indicates that you recently made a request to terminate your Office 365 email and this process has begun by our administrator. If this request was made accidentally and you have no knowledge of it, you are advised to verify your account below [CLICK HERE](#) To verify. Please give us 24 hours to terminate your account OR verify your account. Failure to Verify will result in closure of your account.

<http://offfc4503032.sitebuilder.name.tools/>
Click or tap to follow link.

The link goes to a suspicious website

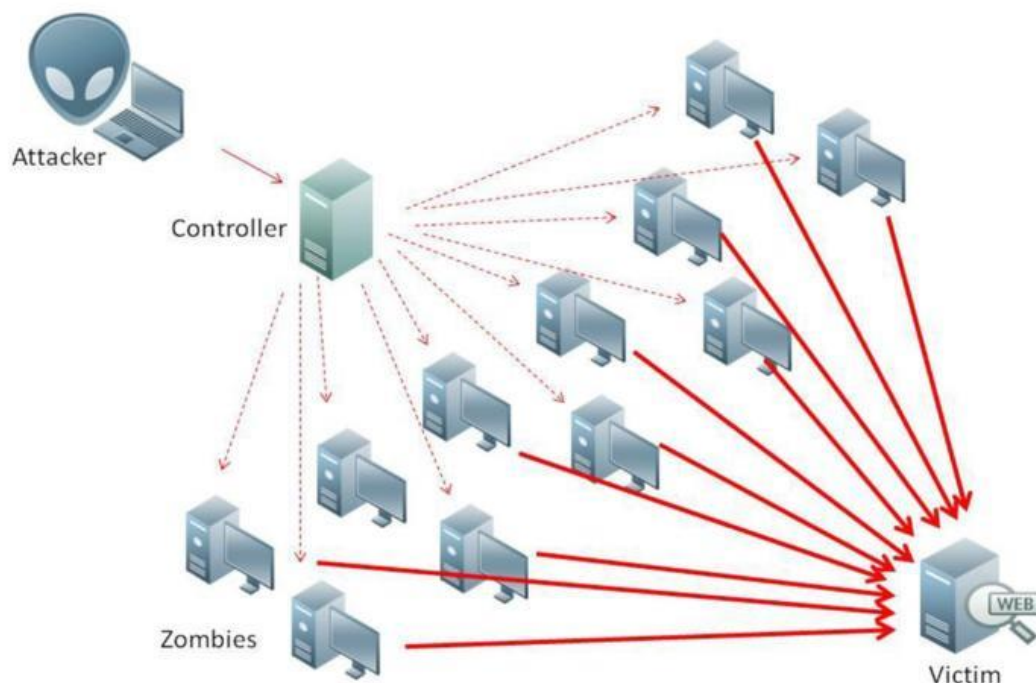
Don't trust an email just because it's from @msvu.ca

IT&S never asks you to click links to verify your account

Watch for spelling, punctuation and grammar errors (highlighted)

DoS (Denial of Service) and DDoS

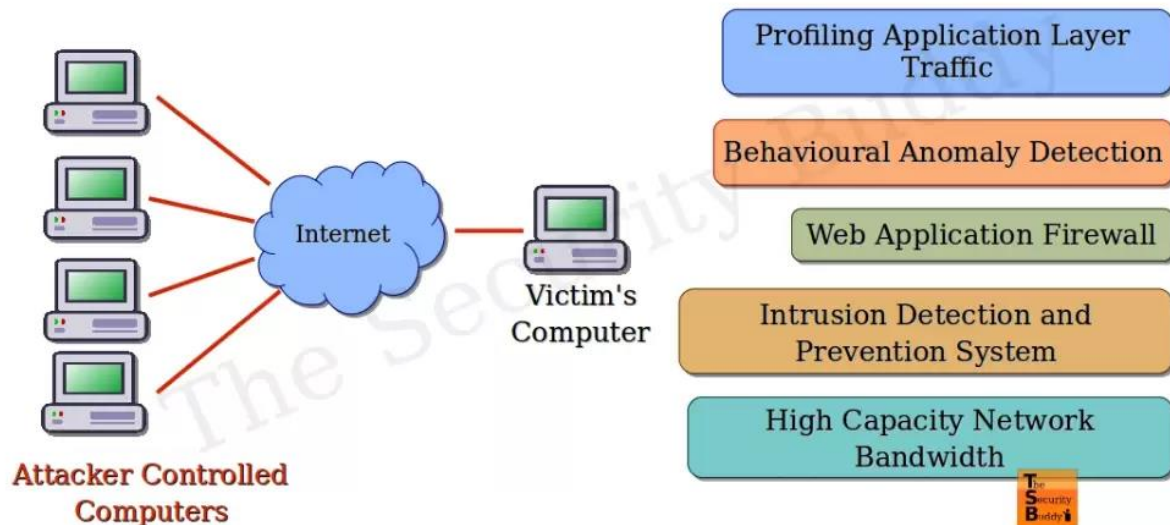
A Denial-of-Service (DoS) attack is an attack meant to shut down a machine or network, making it inaccessible to its intended users. DoS attacks accomplish this by flooding the target with traffic, or sending it information that triggers a crash.



To minimize DoS attack-

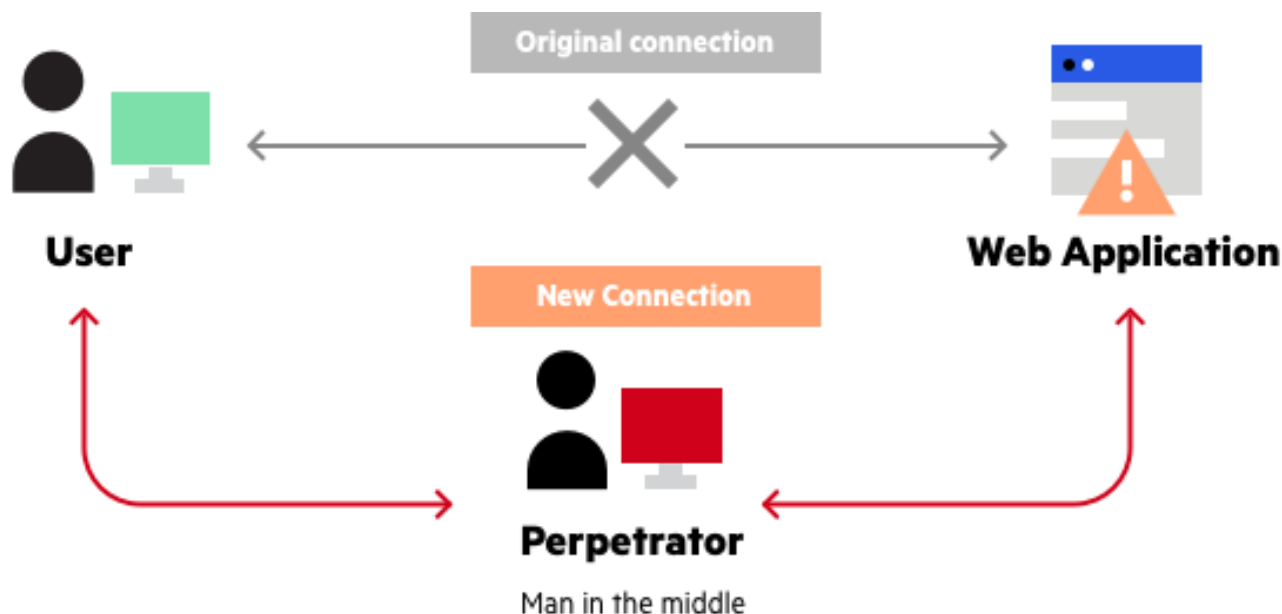
Attack surface reduction: Limiting attack surface exposure can help minimize the effect of a DDoS attack. Several methods for reducing this exposure include restricting traffic to specific locations, implementing a load balancer, and blocking communication from outdated or unused ports, protocols, and applications.

Denial of Service Attacks and Prevention



Man-in-the-middle

A man-in-the-middle (MiTM) attack is a cyberattack where the attacker secretly relays and possibly alters the communications between two parties who believe that they are directly communicating with each other, as the attacker has inserted themselves between the two parties.



Ransomware

- **Ransomware** is a type of [malware](#) that threatens to publish the victim's [personal data](#) or block access to it unless a [ransom](#) is paid.
- While some simple ransomware may lock the system without damaging any files.
- Advanced malware uses a technique called cryptoviral extortion. It [encrypts](#) the victim's files, making them inaccessible, and demands a ransom payment to decrypt them



SQL Injection Attacks

SQL injection, also known as SQLI, is a common attack vector that uses malicious SQL code for backend database manipulation to access information that was not intended to be displayed. This information may include any number of items, including sensitive company data, user lists or private customer details.

